

Shalem Panthagani

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Professional Summary

Software Engineer with 2.5 years of experience building scalable, API-driven systems in Java and SQL. Strong foundation in data structures, algorithms, and problem-solving, with proven success debugging complex production issues and optimizing backend performance under tight SLAs. Skilled in object-oriented design and distributed systems, with additional experience integrating Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) pipelines for intelligent data access and automation solutions.

Skills

Languages: Java, Python, SQL, JavaScript, C, PHP

Frameworks & APIs: Spring Boot, RESTful APIs, JSP/Servlets, GraphQL

Databases: Oracle, MySQL, MariaDB, ChromaDB

Cloud & DevOps: Azure, Bitbucket, Git, CI/CD, Linux, Swagger/OpenAPI, Postman

Data & Analytics: Power BI, sentence-transformers, NLP, Retrieval-Augmented Generation (RAG)

Testing & QA: JUnit, automation scripting, QA processes, Agile/Scrum

System Design & Architecture: Microservices, distributed systems, API architecture, caching, load balancing, performance optimization

Experience

Associate Software Engineer

Temenos AG

Chennai, India

Dec 2021 – Mar 2024

- Reduced recurring client-reported defects by 35% within two quarters by identifying root causes in core transaction modules and implementing long-term fixes, improving uptime and customer satisfaction.
- Automated conversion of proprietary .d/.b records into JSON, removing repetitive manual formatting and improving data consistency across APIs used by multiple internal teams.
- Led migration of 45 repositories from RTC to Bitbucket, defining branching and PR workflows that improved code review turnaround and reduced integration conflicts across teams.
- Improved distributed API response times by resolving latency and throughput bottlenecks in Java-based services, ensuring faster transactions for global banking clients.
- Owned end-to-end defect lifecycle under Temenos's "One-Engineer Model," reproducing, fixing, testing, and releasing independently to ensure faster turnaround and higher delivery reliability.
- Partnered with cross-functional teams to deliver 10+ enhancements for core banking workflows such as payments and reconciliations, improving usability for enterprise operations teams.

Associate Software Engineer

MAQ Software

Hyderabad, India

Apr 2021 – Jul 2021

- Enhanced Power BI dashboards by adding KPIs and visual indicators that improved readability and simplified executive reporting for clients.
- Optimized SQL queries and Power BI data models, reducing data refresh cycles and improving performance during client demos.
- Collaborated with analytics and QA teams in Agile sprints to validate deliverables and ensure timely client releases.

Projects

RAG-Based Support System | *Python, sentence-transformers, ChromaDB, Azure OpenAI*

- Designed and deployed an end-to-end RAG pipeline integrating MiniLM embeddings, token-aware chunking, and ChromaDB for semantic search and retrieval.
- Achieved a 12% improvement in answer accuracy compared to baseline and reduced hallucinations through top-k retrieval and cosine-similarity optimization.

Socratic Debugging Assistant | *Python, LLaMA-3.2, Prompt Engineering, LLM*

- Built an AI-driven assistant using Socratic questioning to guide students in debugging Python code, creating a reproducible framework that improved learning efficiency and supported 100+ student interactions.

Education

Colorado State University - Fort Collins, CO

Master of Computer Information Systems

Aug 2024 – Dec 2025

- Coursework:** Advanced Database Management Systems, Software Product & Process Evaluation, Project Management (IT), Application Software Infrastructure

KL University

Bachelor's in Computer Science

May 2017 – Apr 2021